

SIDNEY, NE, USA

WMO: 725610

Lat: 41.100N

Lon: 102.983W

Elev: 4304

StdP: 12.55

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24030

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-3.3	2.3	-9.7	3.8	-2.2	-3.9	5.2	5.2	37.8	30.3	34.0	28.1	9.6	280	0.700

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.7	97.2	64.2	93.8	63.9	90.8	63.7	69.0	84.8	67.8	84.0	66.5	82.7	12.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.5	106.8	72.6	63.3	102.4	71.7	61.8	96.8	71.7	35.7	84.2	34.5	83.6	33.5	82.5	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
32.4	27.4	24.2	-13.0	101.7	7.4	2.9	-18.3	103.7	-22.7	105.4	-26.8	107.1	-32.2	109.2
			-13.0	71.5	7.5	1.6	-18.3	72.6	-22.7	73.5	-26.9	74.4	-32.3	75.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.2	28.5	29.6	38.8	45.6	55.5	67.3	74.5	72.0	63.3	48.7	37.3	28.2
	DBStd	18.87	10.94	11.65	10.62	9.22	8.68	7.32	5.33	5.56	8.44	9.56	10.28	11.15
	HDD50	3049	666	572	366	186	46	1	0	0	9	136	390	677
	HDD65	6512	1131	991	813	581	311	60	5	13	128	506	831	1142
	CDD50	2765	0	1	18	55	217	519	760	681	408	97	9	0
	CDD65	752	0	0	0	0	17	128	299	229	77	2	0	0
	CDH74	10748	0	0	18	79	436	1867	3868	2916	1365	192	7	0
	CDH80	5210	0	0	0	8	133	854	2103	1453	609	50	0	0
(14) Wind	WSAvg	11.8	12.1	12.6	12.9	13.4	12.0	11.1	10.3	10.3	10.7	11.8	11.8	12.3
(15) Precipitation	PrecAvg	17.2	0.2	0.3	0.7	1.9	2.5	3.1	2.6	2.1	1.5	1.2	0.5	0.3
	PrecMax	24.3	0.7	1.0	2.2	4.3	7.5	8.2	4.9	4.4	5.7	4.5	1.2	1.1
	PrecMin	7.6	0.0	0.0	0.1	0.2	0.2	0.8	1.1	0.2	0.2	0.0	0.0	0.0
	PrecStd	3.7	0.2	0.2	0.6	1.0	1.5	1.8	1.0	1.0	1.2	0.9	0.3	0.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.0	67.9	76.8	81.1	89.6	99.6	101.4	98.9	95.8	85.5	74.3	62.7
		MCWB	43.6	45.7	50.3	53.7	59.7	63.0	64.4	63.5	62.5	56.3	50.4	44.1
	2%	DB	56.2	60.0	70.4	75.7	83.6	93.6	98.1	95.1	91.0	80.0	67.4	55.7
		MCWB	41.0	42.7	48.0	52.1	59.2	63.0	65.0	63.7	60.9	54.8	47.4	40.9
	5%	DB	50.5	53.8	64.4	70.7	79.0	89.4	95.0	91.9	87.6	73.4	61.8	50.3
		MCWB	38.2	39.9	45.7	50.7	58.0	62.9	64.6	63.8	60.2	52.3	44.9	38.2
	10%	DB	45.1	47.6	58.6	65.0	73.8	84.5	91.3	88.7	82.5	67.6	55.0	45.1
		MCWB	35.5	36.7	43.2	48.7	56.1	62.7	65.0	63.9	59.2	50.3	41.9	35.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.0	46.4	52.3	57.1	64.3	69.7	71.1	70.1	66.7	59.2	51.6	44.9
		MCDB	59.6	65.3	71.1	74.5	79.5	83.9	85.8	85.5	84.5	77.6	70.7	60.9
	2%	WB	41.4	43.1	49.1	53.7	61.8	67.6	69.4	68.3	64.4	56.5	48.2	41.6
		MCDB	55.4	58.6	66.9	70.0	76.4	82.5	85.3	84.2	81.2	73.6	66.0	55.2
	5%	WB	38.7	40.3	46.4	51.7	59.8	65.6	68.3	67.1	62.8	54.0	45.5	38.5
		MCDB	49.7	52.7	62.8	68.6	74.0	82.2	84.7	83.0	79.4	69.7	61.0	49.3
	10%	WB	36.0	37.1	44.0	49.6	57.5	64.0	67.1	65.8	61.0	51.4	42.3	35.6
		MCDB	44.5	46.6	57.7	63.6	71.6	80.2	83.7	81.5	77.2	65.5	54.3	44.8
(35) Mean Daily Temperature Range	5% DB	MDBR	25.4	25.0	28.3	27.0	26.5	28.8	29.7	29.4	29.9	28.1	27.1	24.8
		MCDBR	32.6	34.8	38.5	38.4	35.4	36.0	35.8	35.2	37.6	39.3	37.3	32.5
		MCWBR	19.9	20.1	20.3	18.8	15.4	12.7	10.8	11.2	13.7	18.5	20.7	19.9
		MCDBR	30.6	32.9	35.9	34.8	30.4	30.9	29.2	29.0	31.2	33.8	35.7	30.8
	5% WB	MCWBR	19.2	19.5	19.6	18.0	14.4	13.2	10.6	11.0	12.7	17.0	20.1	19.3
(40) Clear-Sky Solar Irradiance	taub		0.246	0.253	0.276	0.301	0.325	0.336	0.364	0.355	0.316	0.279	0.257	0.247
	taud		2.586	2.565	2.521	2.457	2.459	2.455	2.412	2.425	2.511	2.588	2.570	2.559
	Ebn at Noon		299	312	312	308	301	297	287	287	293	297	291	289
	Edh at Noon		21	25	30	34	35	35	36	35	30	24	21	20
(44) All-Sky Solar Radiation	RadAvg		758	1020	1415	1737	2005	2339	2298	2014	1636	1146	819	640
	RadStd		38	60	105	108	155	122	87	71	80	94	47	39

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page